CS1 Midterm Fall 2026

0. Your name:

1. Draw the result of running the following snippet of code. Assume background is white unless specified. Be precise!

a.
```
size(100,100);
line(0,0,width,height);
line(width,0,0,height);
```

b.
```
void setup(){
size(200,200);
}
void draw(){
fill(0);
rect(0,0,width/2,height/2);
rect(width/2,height/2,width/2,height/2);
}
```

2. Translate from Processing into English. For go-getters: can you (slightly) improve the code? Is the condition `x == y` true or false ? Explain.

```
int x=1;
int y=2;
if(x == y){
fill(0,0,0);
rect(0,0,50,50);
}else{
fill(255,0,0);
rect(0,0,50,50);}
```

3. Give short responses to the following questions:

a. what is camelCase?

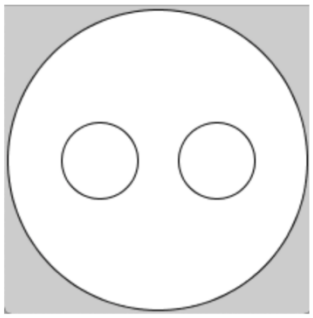
b. What does `x == y` mean?

c. What does `x = y` mean?

d. What are 2 ways to say: set x to 1 less than its current value?

e. What are variables? Why do we need variables of different types? Give examples.

4. Write out code that will produce this image. There are multiple ways to do this so approach it how you would like to.



5. Write a line of code to imitate a throw of a 12-sided die. The result must be an integer between 1 and 12 inclusive.

6. What will be printed out?

```
boolean iLoveProgramming = true;
print("It is definitely "+ !(!(iLoveProgramming)) + " that I love programming!");
```

7. Loops

a. Write a for() loop that will print all even numbers from 6 to 26 (including 26). Rewrite the code using a while() loop.

b. Given code

```
println("tick");
println("tock");
println("tick");
println("tock");
println("tick");
println("tock");
```

Rewrite using a loop.

8. Variables —watch out, might be tricky!
    a. What is a variable scope?

    b. You wrote a snippet of code:

```
 int x = 20;
 void setup(){
size(200,200);
x = 200;
 int y = 50;
}
 void draw(){
  println(x/y);
}
```

Will this code compile? If not, why? What will be printed to the console? Why?

c. You wrote a snippet of code:

int a=1;

a--; //what is a now?

a-=4; //what is a now?

int b=a;//what is b now?

Draw a memory diagram.

9. Calling a function

```
a.  int p(int n){
     int prod=1;
     for(int i=0;i<n;i++){
       prod*=2;
     }
     return prod;
```

```
}
```

What does the function above return if n=2? Show a memory diagram.
What does the function above return if n=4? Show a memory diagram.
Give a one-sentence "big picture" description of the function or express the output as a function of n.

10. Arrays

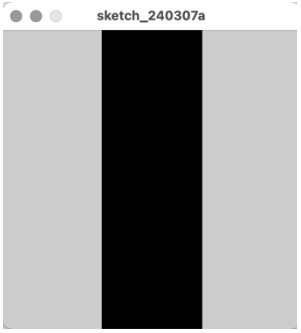
Your first-year friend doesn't know what major to pick but they've narrowed down their choices.

```
//one of multiple ways to declare a String array
String[] majors ={"Math and CS", "Biology", "Graphic Design",
"Creative Writing", "New Media", "Music: Performance", "Theatre
and Performance"};

println(majors);
```

The above starter code prints the list of majors they are considering. - Now add a line of code to pick a major at random from their list and print it to the console. (Note: this is probably not a recommended way to pick a major.)

11. Write the condition that the mouse is in the black area; the gray area.



12. Consider a snippet of code. What will be printed on the console? Explain. Again, watch out, might be tricky!

```
int i;
for(i=10; i<14;i++);{
println(i);
}
```

13.(**for go-getters**) Consider a snippet of code. What will be printed to the console?

```
int []arr={1,2,3,4,5,6,3,2,1,0};
 println(arr[4]);
 println(arr[arr[4]]);
 println(arr[arr[3]+arr[9]]);
 println(arr[arr[3]]+arr[arr[5]]);
```

You added a line of code

```
println(arr[arr[3]+arr[5]]);
```

What will happen now?